Realcryptik: Pragmatic Election Verification with Zeeperio

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Abstract. The size of a Zeeperio proof is constant in the number of ballots and voters in an election, as is the time to verify it. It is compact enough for autonomous verification by a smart contract on a blockchain, such as Ethereum.

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1 Introduction

Decades of research on end-to-end verifiable (E2E-V) elections have shown strong cryptographic guarantees can be realized for protecting the integrity of elections and the secrecy of voter ballots [16]. In more recent years, pilot projects and deployments of E2E-V platforms have also demonstrated that these systems are usable and scalable to an appropriate degree [2, 8, 4, 12]. Despite these efforts, the vendors of systems for governmental elections continue to run unverifiable elections where the content of individual ballots can be determined by at least one trusted entity in the platform.

In this paper, we contribute to a broader discussion the cryptographic voting community has been having for some number of years: do we compromise our ideals and offer watered down systems that the industry might find more palatable if they offer tangible security advantages over the status quo? We relate this to the term Realpolitik, coined in 19th century Germany to describe statecraft based on pragmatic calculations rather than ideology, and propose *Realcryptik* to describe such a pragmatic approach to election verification.

Flavours of Realcryptik already permeate the research literature. Despite opposition to internet voting, Helios [1] is perhaps the most used E2E-V voting technology. While researchers argue coercion resistance is essential for internet voting, Helios provides none. Scantegrity II [9] was the first E2E-V system to see governmental use. It is built around the compromise that in a paper ballot system, it is ok for optical scanner to learn how the voter voted, side-stepping the complex ballot mechanics its predecessors to avoid this. In recent years, serious

discussions have arose around whether E2E-V should be largely abandoned in favour of risk limiting audits? Should we affirm the use of ballot marking devices when we know most voters do not carefully check their output?

We propose a different compromise in this paper: let EAs and their vendors see the raw ballots, while protecting them from all outsiders (receipt-freeness). For the EA, we still rely on legal protections implemented through procedures but not offer cryptographic protection. Should add more arguments...

The research question of this paper is how to modify the cryptography of an E2E-V voting system to provide integrity, without the need to maintain ballot secrecy from the EA (but still provide receipt-freeness otherwise). The simpler threat model means we can streamline the cryptography, in particular we can prove that encrypted (or committed in our case) ballots are tallied correctly rather than computing the tally under encryption. This shifts the protocol away from multiparty computation and threshold cryptography to essentially an all encompassing zero knowledge proof (ZKP). While ZKPs have been used in voting systems from almost their invention in 1980s and onward, we focus on modern succinct ZKPs called zk-SNARKs.

Zeeperio. We apply this modification to an existing E2E-V back-end called Eperio [13]. Eperio is a simplification of Scantegrity II's backend (known as a switchboard) and is design to be simple enough that a non-expert validator can audit the election using a spreadsheet, such as MS Excel. Eperio's security was studied in detail using formal methods at CCS 2021 paper [3]. By replacing Eperio's cut-and-choose based proof with a zk-SNARK, it obviously is no longer verifiable by hand. However this is a deliberate choice. Traditionally voters would either have to verify an election themselves or rely on a trusted entity (political party, watchdog) to perform this verification—hence the need to simplify elections to a spreadsheet. Today we have systems like Ethereum that provide a peer-to-peer network that is effectively validating that computational outputs of arbitrary code, called smart contracts, are correct. Ethereum protects assets with a market capitalization of $100+B USD and it can protect election results too, as long as the results are verifiable within its very restricted limits (known as gas consumption). If the verification task can be reduced to a succinct (zk-SNARK) proof, voters do not need check the proof at all.

Our contribution is designing a custom succinct argument for Eperio that we call Zeeprio. Our protocol uses polynomial interactive oracle proofs (PIOPs), which have been used to build zk-SNARKs for any arithmetic circuit (and thus any computation), the best known system being Plonk [14]. The size of a Zeeperio proof is constant in the number of ballots and voters in an election, as is the time to verify it. It is compact enough for autonomous verification by a smart contract on a blockchain, such as Ethereum. Zeeperio proofs are linear (as opposed to constant) in the number of candidates, but this seems acceptable as reading the final tally or checking an audited ballot is linear in the number of candidates.

2 Background and Related Work

2.1 Eperio

We assume the reader is already familiar with end-to-end verifiable (E2E-V) voting protocols which allow for: (i) an election authority to prove an election tally is computed correctly without learning individual voter choices, and for (ii) voters to check their individual ballots are included unmodified in the tally without being able to reveal how they voted to anyone else. Eperio is an E2E-V tallying protocol (or 'backend') that composes with several styles of paper ballots, including Pret a Voter and Scantegrity I. A brief description of Eperio is provided in Figure **??** with a Pret a Voter style ballot.

In addition to succinctness, we also clean up a few issues that were not fully resolved in Scantegrity II and Eperio:

- As pointed about by Basin et al, Eperio does not prove ballots were not overvoted [3]; Zeeperio does (§**??**).
- As acknowledged by Essex et al, Eperio does not support Scantegrity II ballots [13]; Zeeperio does (§**??**).
- As acknowledged by Chaum et al, Scantegrity II supports dispute resolution but it requires all confirmation codes on the ballot to be revealed [9, 10]. Zeeprio offers a simple proof of incorrect code without revealing information about all codes (§**??**).

A remaining issue, pointed out by Basin et al [3], are clash attacks [17] which are not explicitly addressed by Eperio. However they are applicable to all major E2E-V voting systems and are addressed through strict polling place policies that denote which ballot is given to which voter.

2.2 Succinct Proofs

While zero knowledge proofs/arguments (ZKPs) date back to the 1980s [15, 6], classic ZKPs produce proofs that are at least linear in the size of the statement being proven. By contrast, succinct zero knowledge arguments (SNARKs) are sublinear, typically with a logarithmic or constant proof size. A Polynomial Interactive Oracle Proof (PIOPs) is a pattern that a number of SNARKs have in common [11, 7]. In this pattern, the prover specifies polynomials and verifiers challenge the prover to open these polynomials at random points. PIOP does not say *how* to make a SNARK, it is just a general pattern.

To realize a PIOP, the prover must send a succinct representation of each polynomial from the prover to the verifier, which commits the prover to those polynomials. When opening a polynomial, the verifier must be certain it is the opening of the same polynomial that was committed to. PIOPs rely on a cryptographic primitive called a polynomial commitment scheme (PCS) to handle this. The rest of a PIOP protocol does not use any cryptography. A number of PCSs have been proposed, the most prominent are KZG (very short, based on

bilinear pairings, requires a trusted setup) and (DEEP-)FRI (longer, based on hash functions, conjectured to be post-quantum secure, no trusted setup). In Zeeperio, we use KZG because it integrates well with Ethereum.

KZG uses a universal trusted setup: a secret value τ is chosen at random, used to compute a structured reference string (SRS) $\left\langle g^{\tau}, g^{\tau^2}, g^{\tau^3}, \ldots \right\rangle$ in a pairing-friendly group defined by generator g, and then erased. Knowledge of τ after the setup is sufficient to fake KZG openings and thus forge false ZKP transcripts. However the setup can be run as an open participation multiparty computation where participants prove their contribution is correct and only one honest participant must forget τ for the SRS to be secure. Ethereum ran an open trusted setup in 2023, accepting 100K contributions over 208 days. A single SRS, such as this, can be shared by everyone wanting to do a KZG commitment (known a universal trusted setup).

In practice, zk-SNARKs for a specific computation can be deployed in different ways: (a) written in a high-level language and compiled to VM instructions, (b) written in special markup languages and compiled into arithmetic circuits, or (c) written as a custom arithmetic circuit. In each case, these representations are fed into a general purpose proof system for circuits or zkVMs. Each level of abstraction adds overhead and costs to the prover. Our approach is 'closer to the metal.' We do not use circuits or Plonk itself. We instead go underneath what Plonk provides and use the same kind of building blocks it uses to make a custom PIOP argument for Zeeperio.

2.3 Smart Contracts

Smart contracts are programs deployed on a blockchain, such as Ethereum. They only execute when a user calls a function. The blockchain's consensus enforces the function call executes correctly and the state of the code is updated accordingly. In Zeeprio, a smart contract will accept an election proof and run the verification of it, reporting if it is correct or incorrect. Since smart contracts can implement arbitrary code, they can support cryptographic operations however this approach is usually prohibitively expensive. Ethereum offers cheaper 'pre-compiles' for pairing-based operations on the elliptic curve `alt-bn128`. On Ethereum's testnet, support for curve `bls12-381` has been staged for future deployment, as well as KZG-specific operations over `bls12-381`, using the forementioned SRS. Zeeperio on `alt-bn128` or `bls12-381`

3 Zeeperio

Parameters. Let $\mathcal{B}$ be the number of printed ballots and $\mathcal{C}$ the number of candidates per ballot. Let $\mathcal{P} = \mathcal{BC}$ be the number of ballot positions. Let n be the padded column length (a power of two) with $n \geq \mathcal{P}$ (the need is explained below). Let $\mathbb{F}_q$ be a finite field of size q. A ballot has ID $b \in \{0, \ldots, \mathcal{B}-1\}$ and occupies the *ballot block*:

$$I_b = \{\, b\mathcal{C},\, b\mathcal{C}+1,\, \ldots,\, b\mathcal{C}+(\mathcal{C}-1) \,\} \subseteq \{0, \ldots, n-1\}.$$

Walkthrough of a constraint. Notice from Figure **??**, the mark column can only contain a 0 or 1. This is one small fact about what constitutes a valid election table. With ZKPs, we call this a *constraint*. Zeeperio consists of 16 similar constraints that all must hold in any valid election table. Here we will demonstrate a complete argument for this single constraint (which we denote as c2.2 later) to demonstrate how a PIOP works. This constraint is adapted from [5].

The mark column can be considered a one-dimensional vector or array $A_{\mathsf{mark}}[i]$ of length $\mathcal{P}$. Elements in the array will be encoded as elements of $\mathbb{F}_q$, the exponent field of a pairing-friendly elliptic curve such as `alt-bn128` or `bls12-381`. Cryptographically speaking, the array $A_{\mathsf{mark}}[i]$ could thus hold any integer in $[0, q-1]$ but the constraint says it is only a valid election table if it is further restricted to $[0, 1]$.

To prove the constraint, the prover (election authority) will send to the verifier (smart contract) a succinct (one group element) commitment (hiding and binding) to the entire mark column $A_{\mathsf{mark}}[i]$. While vector commitments exist, in a PIOP, the prover commits to $A_{\mathsf{mark}}[i]$ indirectly by first encoding the data from $A_{\mathsf{mark}}[i]$ into a univariate polynomial over $\mathbb{F}_q$ and then commits to this polynomial representation of $A_{\mathsf{mark}}[i]$ using a polynomial commitment scheme such as KZG.

There a variety of ways to encode vectors into polynomials (we refer interested reader to plonkbook), we will focus on the method we will use. The prover chooses a set of x-coordinates, such as $\{0, 1, 2, 3, \ldots\}$ which is called an evaluation domain $\mathcal{H}$. It then takes the data from $A_{\mathsf{mark}}[i]$ and forms a set of points $\{(0, A_{\mathsf{mark}}[0]), (1, A_{\mathsf{mark}}[1]), \ldots\}$. The prover interpolates a univariate polynomial through these points: $\mathsf{Mark}(X)$. It commits to $\mathsf{Mark}(X)$ using the polynomial commitment scheme, and sends the commitment to the verifier.

Interpolation is expensive for the prover so provers use an optimization. The evaluation domain is arbitrary and can be anything as long as the prover and verifier agree. Let $\omega \in \mathbb{F}_q$ generate a multiplicative subgroup of size n. Then set $H = \{\omega^0, \omega^1, \ldots, \omega^{n-1}\} \subseteq \mathbb{F}_q$. Doing so offers three advantages and one drawback. Interpolating a polynomial over this domain can be accomplished in $\mathcal{O}(n \log n)$ instead of $\mathcal{O}(n^2)$. Traversing the domain easy: the next index is the current index multiplied by ω and because it is closed under multiplication, the 'next' element after the last element ω^{n-1} wraps back to the first element $\omega^{n-1} \cdot \omega = \omega^n = \omega^0$. Finally, a polynomial that is 0 at each $X \in H$ has a simple closed form: $\mathsf{Z}_H(X) = X^n - 1$. The drawback is that the prover cannot use arrays of arbitrary length $\mathcal{P}$ and expect that an order n subgroup will just exist. The subgroups are based on the parameters of the elliptic curve and curves like `alt-bn128` and `bls12-381` are designed to offer subgroups of sizes $\{2, 2^2, 2^3, 2^4, \ldots\}$ up to 2^{12} for `alt-bn128` and 2^{32} for `bls12-381`.

In Zeeprio, if we have a column of length $\mathcal{P}$, we will zero pad it the nearest power of 2. Thus n is the padded column length (a power of two) with $n \geq \mathcal{P}$. For this example constraint (mark column is binary), the presence of zero padding does not impact the constraint. However for other constraints, we will have to explicitly handle the padding separately from the election table data.

At this point, the verifier has a polynomial commitment to $\mathsf{Mark}(X)$ which we will call $K_{\mathsf{Mark}(X)}$. How does the prover demonstrate it is binary? The prover will construct another polynomial as follows: $V(X) := \mathsf{Mark}(X)\big(\mathsf{Mark}(X)-1\big)$. It will commit to $V(X)$ and send $K_{V(X)}$ to the verifier. If $\mathsf{Mark}(X)$ is binary for all $X \in H$, then $V(X)$ will be zero for all $X \in H$. If $\mathsf{Mark}(X)$ is not binary for at least one $X \in H$, $V(X)$ will be non-zero at the same X. We have converted the constraint $\mathsf{Mark}(X)$ is binary into:

1. $V(X)$ is correctly constructed from $\mathsf{Mark}(X)$
2. $V(X)$ it is zero for all $X \in H$. A polynomial that is is zero for all $X \in H$ is called a *vanishing polynomial* (on H).

To prove that $V(X)$ is correctly constructed from $\mathsf{Mark}(X)$, the verifier has $K_{\mathsf{Mark}(X)}$ and $K_{V(X)}$. The verifier will (or the prover using the strong Fiat-Shamir heuristic) choose a random evaluation point $\zeta \notin H$. Using the polynomial commitment scheme, the prover will open the polynomials at the value ζ sending $V(\zeta)$, $\mathsf{Mark}(\zeta)$, and opening proofs for each to the verifier. The verifier will check the opening proofs and check that $V(\zeta) := \mathsf{Mark}(\zeta)\big(\mathsf{Mark}(\zeta)-1\big)$. Just because the relation holds at the point ζ, what does that say about it holding at each $X \in H$? The answer is that it overwhelmingly holds due to a lemma named Schwartz-Zippel that we will not review further here.

Next the prover will demonstrate that $V(X)$ vanishes on H. Recall that $\mathsf{Z}_H(X) = X^n - 1$ is a vanishing polynomial on H. The prover will construct another polynomial $Q(X) := V(X)/\mathsf{Z}_H(X)$, commits to $Q(X)$, and sends the commitment $K_{Q(X)}$ to the verifier. If and only if $V(X)$ is zero at every point of H, $\mathsf{Z}_H(X)$ exactly divides $V(X)$ and results in a proper polynomial $Q(X)$, otherwise $Q(X)$ will be a rational function not reducible to a polynomial. $Q(X) := V(X)/\mathsf{Z}_H(X)$ can be rearranged as $V(X) - Q(X) \cdot (X^n - 1) = 0$. Using ζ from above, the prover sends $Q(\zeta)$ and its opening proof. The verifier checks $V(\zeta) - Q(\zeta) \cdot (\zeta^n - 1) = 0$. By Schwartz–Zippel, if this equality holds at a uniformly random $\zeta \notin H$, then except with negligible probability it holds as a polynomial identity, which implies that $V(X)$ vanishes on H. □

3.1 Zeeperio Constraints

Zeeperio is made up of 16 constraints. Each constraint follows the similar pattern to the example constraint: a set of polynomials are committed to such that a relationship between them will be a vanishing polynomial if and only if the constraint is true. Once the constraint is converted into a vanishing polynomial, proving the vanishing polynomial is constructed correctly and that it actually vanishes is identical to the process used above.

In fact, given a set of vanishing polynomials $\{V_1(X), V_2(X), \ldots, V_m(X)\}$ and their corrosponding quotiant polynomials, the check can be batched into a single check. The prover will form a 'super-polynomial' $\mathsf{Batch}(X)$ where each coefficient is one of the vanishing polynomials:

Table 1. A set of polynomials that can be computed before the beginning of the election. $\mathsf{Sel}_b(X)$ is the one exception, it will not be precomputed but rather computed during a receipt check just for the one ballot b that the voter wants to check.

Description	**Polynomial**
Polynomial that is zero on H	$\mathsf{Z}_H(X) = X^n - 1$
Mask that is 0 on all of H except possibly at $X = \omega^0$	$\mathsf{Mask}_0(X) = \dfrac{\mathsf{Z}_H(X)}{X - \omega^0}$
Mask that is 0 on all of H except possibly at $X = \omega^{n-1}$	$\mathsf{Mask}_{n-1}(X) = \dfrac{\mathsf{Z}_H(X)}{X - \omega^{n-1}}$
Polynomial that is 0 up to padding $[0, \mathcal{P} - 1]$	$\mathsf{Z}_{\mathsf{head}}(X) = \prod_{i=0}^{\mathcal{P}-1} \left(X - \omega^i\right)$
Polynomial that is 0 for the last $\mathcal{C}$ rows (zeros at $\omega^{n-1}, \ldots, \omega^{n-\mathcal{C}}$)	$\mathsf{Mask}_{\mathsf{tail}}(X) = \prod_{t=0}^{\mathcal{C}-1} \left(X - \omega^{n-1-t}\right)$
Lagrange basis over H used below	$L_i(X) = \prod_{\substack{0 \le j < n \\ j \ne i}} \dfrac{X - \omega^j}{\omega^i - \omega^j}$
Polynomial with a 1 at the start of every ballot (indices $i \equiv 0 \pmod{\mathcal{C}}$ up to padding) and 0 elsewhere	$\mathsf{Sel}_{\mathsf{blk}}(X) = \sum_{j=0}^{\lfloor n/\mathcal{C} \rfloor - 1} L_{j\mathcal{C}}(X)$
Polynomial that is 1 for all rows belonging to ballot-b ($I_b = \{b\mathcal{C}, \ldots, b\mathcal{C} + \mathcal{C} - 1\}$) and 0 elsewhere	$\mathsf{Sel}_b(X) = \sum_{k=0}^{\mathcal{C}-1} L_{b\mathcal{C}+k}(X)$

$$\mathsf{Batch}(Y) = \sum_{i=1}^{m} Y^i \Big(V_i(X) - Q_i(X)\, \mathsf{Z}_H(X) \Big)$$

A polynomial where every coefficient is 0 is a zero-polynomial. Therefore the verifier (or prover via strong Fiat-Shamir) will choose a random $\zeta \in \mathbb{F}_q \notin H$ as before and set $X = \zeta$, which makes every coefficient zero. To see if every coefficient is zero, it will also choose a random $\alpha \in \mathbb{F}_q$ and set $Y = \alpha$. If every $V_i(X)$ vanishes, then $\mathsf{Batch}(Y)$ is a zero polynomial and:

$$\sum_{i=1}^{m} \alpha^i \Big(V_i(\zeta) - Q_i(\zeta)\, \mathsf{Z}_H(\zeta) \Big) = 0.$$

3.2 Public Polynomials

Zeeperio uses several public helper polynomials that depend only on the election 'shape' (e.g., n, $\mathcal{B}$, $\mathcal{C}$, and hence $\mathcal{P} = \mathcal{BC}$), and not on any voter choices or on the contents of the committed columns. They are enumerated in Table 1. Some of these helpers are linear or superlinear in n to construct (for example, explicit products or explicit sums of Lagrange basis polynomials). This does not undermine the succinctness claims of Zeeprio: the pre-computation is offline, public,

and independent of the witness. In systems like Plonk, these types of polynomials are considered as a 'verifier's key.' In practice, election administrators can standardize on a small set of template sizes and these polynomials can be precomputed once for each election template. If an election uses fewer than the template's maximum number of ballots or candidates, the remaining rows are constrained to be empty: extra ballots are unused and extra candidates receive zero votes, which is directly reflected in the published tally.

3.3 Election Specific Polynomials

The Eperio tables for an election consist of three (padded) columns which will be encoded into polynomials evaluated over H. We call the polynomials $\mathsf{Code}(X)$, $\mathsf{Audit}(X)$, and $\mathsf{Mark}(X)$. How each index maps to a unique ballot ID and candidate is provided as a public template and columns will be sorted by ballot ID and then candidate (see Figure **??**). The final tally is a vector $\mathsf{Tally}[0..\mathcal{C}-1] \in \mathbb{F}_q^{\mathcal{C}}$.

Across all the constraints, we will use an additional 5 'helper' polynomials. We will introduce them as needed, however they consist of 3 accumulator polynomials: $\mathsf{Acc}_{\mathsf{Audit}}(X)$,$\mathsf{Acc}_{\mathsf{Mark}}(X)$, and $\mathsf{Acc}_{\mathsf{Vote}}(X)$. As well as two block sum helper polynomials: $\mathsf{BlkSum}_{\mathsf{Audit}}(X) = \sum_{k=0}^{\mathcal{C}-1} \mathsf{Audit}(\omega^k X)$ and $\mathsf{BlkSum}_{\mathsf{Mark}}(X) = \sum_{k=0}^{\mathcal{C}-1} \mathsf{Mark}(\omega^k X)$.

4 Voter Checks

4.1 Print Audit

Indices and ballots. We index rows by $i \in \{0, \ldots, n-1\}$ and identify the i-th domain point with $X = \omega^i$. A ballot has ID $b \in \{0, \ldots, \mathcal{B}-1\}$ and occupies the *ballot block*

$$I_b = \{\, b\mathcal{C},\, b\mathcal{C}+1,\, \ldots,\, b\mathcal{C}+(\mathcal{C}-1) \,\} \subseteq \{0, \ldots, n-1\}.$$

Block selector for ballot b. Let $\mathsf{Sel}_b(X)$ be the (publicly checkable) selector polynomial for ballot b defined pointwise by

$$\mathsf{Sel}_b(\omega^i) = \begin{cases} 1, & i \in I_b, \\ 0, & \text{otherwise}, \end{cases} \qquad i \in \{0, \ldots, n-1\}.$$

(Equivalently, Sel_b is the interpolation of this 0/1 array over H.)

c5.1 Print-audit check. The voter supplies ballot ID b. The EA returns the $\mathcal{C}$ evaluations

$$\big\{\mathsf{Audit}(\omega^i)\big\}_{i \in I_b}, \qquad \big\{\mathsf{Mark}(\omega^i)\big\}_{i \in I_b}, \qquad \big\{\mathsf{Code}(\omega^i)\big\}_{i \in I_b},$$

together with a batched opening proof for these evaluations.

Table 2. Audit column constraints (c1.*).

c1.1 Audit is zero-padded. For all $X \in H$: $\mathsf{Audit}(X) \cdot \mathsf{Z}_{\mathsf{head}}(X) = 0$.

$$V_{c1.1}(X) \;:=\; \mathsf{Audit}(X)\,\mathsf{Z}_{\mathsf{head}}(X)$$

c1.2 Audit is binary. For all $X \in H$: $\mathsf{Audit}(X)\big(\mathsf{Audit}(X) - 1\big) = 0$.

$$V_{c1.2}(X) \;:=\; \mathsf{Audit}(X)\big(\mathsf{Audit}(X) - 1\big)$$

c1.3 Audit is block-consistent: each ballot block is all-0 or all-1. For all $X \in H$: $\big(\mathsf{Sel}_{\mathsf{blk}}(X)\mathsf{BlkSum}_{\mathsf{Audit}}(X)\big)\big(\mathsf{Sel}_{\mathsf{blk}}(X)\mathsf{BlkSum}_{\mathsf{Audit}}(X) - \mathcal{C}\big) = 0$, where $\mathsf{BlkSum}_{\mathsf{Audit}}(X) = \sum_{k=0}^{\mathcal{C}-1} \mathsf{Audit}(\omega^k X)$.

$$V_{c1.3}(X) \;:=\; \big(\mathsf{Sel}_{\mathsf{blk}}(X)\mathsf{BlkSum}_{\mathsf{Audit}}(X)\big)\big(\mathsf{Sel}_{\mathsf{blk}}(X)\mathsf{BlkSum}_{\mathsf{Audit}}(X) - \mathcal{C}\big)$$

c1.4a Audit accumulator step (suffix sum). For all $X \in H \setminus \{\omega^{n-1}\}$: $\mathsf{Acc}_{\mathsf{Audit}}(X) - \mathsf{Audit}(X) + \mathsf{Acc}_{\mathsf{Audit}}(\omega X) = 0$.

$$V_{c1.4a}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Audit}}(X) - \mathsf{Audit}(X) + \mathsf{Acc}_{\mathsf{Audit}}(\omega X)\big) \cdot (X - \omega^{n-1})$$

c1.4b Audit accumulator base (last row). For $X = \omega^{n-1}$: $\mathsf{Acc}_{\mathsf{Audit}}(X) - \mathsf{Audit}(X) = 0$.

$$V_{c1.4b}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Audit}}(X) - \mathsf{Audit}(X)\big) \cdot \mathsf{Mask}_{n-1}(X)$$

c1.4c Audit sum matches public count. For $X = \omega^0$: $\mathsf{Acc}_{\mathsf{Audit}}(X) - \mathsf{Sum}_{\mathsf{Audit}\times\mathcal{C}} = 0$.

$$V_{c1.4c}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Audit}}(X) - \mathsf{Sum}_{\mathsf{Audit}\times\mathcal{C}}\big) \cdot \mathsf{Mask}_0(X)$$

4.2 Receipt Check

c5.2 Receipt check (shuffle-based). Ballot-ID column. Let $\mathsf{BallotID}(X)$ be a public column encoding ballot IDs per position, defined by

$$\mathsf{BallotID}(\omega^{b\mathcal{C}+k}) = b \quad \text{for } b \in \{0,\ldots,\mathcal{B}-1\},\ k \in \{0,\ldots,\mathcal{C}-1\},$$

and $\mathsf{BallotID}(\omega^i) = 0$ on padding indices $i \geq \mathcal{P}$.

Secret permutation. Let π be a secret permutation of $\{0,\ldots,n-1\}$. For any column F, define its permuted form

$$\mathsf{F}^{\pi}(\omega^i) = \mathsf{F}\big(\omega^{\pi(i)}\big).$$

The EA commits once (pre-election or once-per-election) to

$$\mathsf{BallotID}^{\pi}(X), \quad \mathsf{Code}^{\pi}(X), \quad \mathsf{Mark}^{\pi}(X),$$

Table 3. Mark column constraints (c2.*).

c2.1 Mark is zero-padded. For all $X \in H$: $\mathsf{Mark}(X) \cdot \mathsf{Z}_{\mathsf{head}}(X) = 0$.

$$V_{c2.1}(X) \;:=\; \mathsf{Mark}(X)\,\mathsf{Z}_{\mathsf{head}}(X)$$

c2.2 Mark is binary. For all $X \in H$: $\mathsf{Mark}(X)\big(\mathsf{Mark}(X) - 1\big) = 0$.

$$V_{c2.2}(X) \;:=\; \mathsf{Mark}(X)\big(\mathsf{Mark}(X) - 1\big)$$

c2.3 Mark has at most one '1' per ballot block. For all $X \in H$: $\big(\mathsf{Sel}_{\mathsf{blk}}(X)\mathsf{BlkSum}_{\mathsf{Mark}}(X)\big)\big(\mathsf{Sel}_{\mathsf{blk}}(X)\mathsf{BlkSum}_{\mathsf{Mark}}(X) - 1\big) = 0$, where $\mathsf{BlkSum}_{\mathsf{Mark}}(X) = \sum_{k=0}^{\mathcal{C}-1} \mathsf{Mark}(\omega^k X)$.

$$V_{c2.3}(X) \;:=\; \big(\mathsf{Sel}_{\mathsf{blk}}(X)\mathsf{BlkSum}_{\mathsf{Mark}}(X)\big)\big(\mathsf{Sel}_{\mathsf{blk}}(X)\mathsf{BlkSum}_{\mathsf{Mark}}(X) - 1\big)$$

c2.4a Mark accumulator step (suffix sum). For all $X \in H \setminus \{\omega^{n-1}\}$: $\mathsf{Acc}_{\mathsf{Mark}}(X) - \mathsf{Mark}(X) + \mathsf{Acc}_{\mathsf{Mark}}(\omega X) = 0$.

$$V_{c2.4a}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Mark}}(X) - \mathsf{Mark}(X) + \mathsf{Acc}_{\mathsf{Mark}}(\omega X)\big) \cdot (X - \omega^{n-1})$$

c2.4b Mark accumulator base (last row). For $X = \omega^{n-1}$: $\mathsf{Acc}_{\mathsf{Mark}}(X) - \mathsf{Mark}(X) = 0$.

$$V_{c2.4b}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Mark}}(X) - \mathsf{Mark}(X)\big) \cdot \mathsf{Mask}_{n-1}(X)$$

c2.4c Mark sum matches public count. For $X = \omega^0$: $\mathsf{Acc}_{\mathsf{Mark}}(X) - \mathsf{Sum}_{\mathsf{Mark}} = 0$.

$$V_{c2.4c}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Mark}}(X) - \mathsf{Sum}_{\mathsf{Mark}}\big) \cdot \mathsf{Mask}_0(X)$$

and proves (once) that each is a permutation of its source column under the *same* π.

Per-receipt query. Given ballot ID b, the EA finds (privately) an index t such that

$$\mathsf{BallotID}^{\pi}(\omega^t) = b \quad \text{and} \quad \mathsf{Mark}^{\pi}(\omega^t) = 1,$$

and returns the claimed receipt code

$$c := \mathsf{Code}^{\pi}(\omega^t),$$

along with a batched opening proof at the single point ω^t for $\mathsf{BallotID}^{\pi}, \mathsf{Mark}^{\pi}, \mathsf{Code}^{\pi}$. (Revealing t is safe because π is secret, so t carries no candidate-order meaning.)

Table 4. Cross-column constraint (c3).

c3 No ballot is both audited and voted. For all $X \in H$: $\mathsf{Audit}(X) \cdot \mathsf{Mark}(X) = 0$.

$$V_{c3}(X) \;:=\; \mathsf{Audit}(X)\,\mathsf{Mark}(X)$$

Table 5. Tally constraints (c4.*).

c4.1 Strided vote-accumulator step (stride $\mathcal{C}$). For all $X \in H \setminus \{\omega^{n-1}, \ldots, \omega^{n-\mathcal{C}}\}$: $\mathsf{Acc}_{\mathsf{Vote}}(X) - \mathsf{Mark}(X) + \mathsf{Acc}_{\mathsf{Vote}}(\omega^{\mathcal{C}} X) = 0$.

$$V_{c4.1}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Vote}}(X) - \mathsf{Mark}(X) + \mathsf{Acc}_{\mathsf{Vote}}(\omega^{\mathcal{C}} X)\big) \cdot \mathsf{Mask}_{\mathsf{tail}}(X)$$

c4.2 Strided vote-accumulator tail base. For all $X \in \{\omega^{n-1}, \ldots, \omega^{n-\mathcal{C}}\}$: $\mathsf{Acc}_{\mathsf{Vote}}(X) - \mathsf{Mark}(X) = 0$.

$$V_{c4.2}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Vote}}(X) - \mathsf{Mark}(X)\big) \cdot \frac{\mathsf{Z}_H(X)}{\mathsf{Mask}_{\mathsf{tail}}(X)}$$

c4.3 Tally readout. For each $c \in \{0, \ldots, \mathcal{C}-1\}$, for $X = \omega^c$: $\mathsf{Acc}_{\mathsf{Vote}}(X) - \mathsf{Tally}[c] = 0$.

$$V_{c4.3,c}(X) \;:=\; \big(\mathsf{Acc}_{\mathsf{Vote}}(X) - \mathsf{Tally}[c]\big) \cdot \frac{\mathsf{Z}_H(X)}{X - \omega^c}$$

4.3 Dispute Resolution

c5.3 Dispute resolution (selector + equality flags). The voter supplies ballot ID b and disputed code $c \in \mathbb{F}_q$. The EA proves that c does not appear on that ballot block without revealing any codes.

Equality flag polynomial. The EA introduces a binary flag polynomial $\mathsf{Eq}(X) \in \{0, 1\}$ over H intended to satisfy

$$\mathsf{Eq}(\omega^i) = \begin{cases} 1, & \mathsf{Code}(\omega^i) = c, \\ 0, & \text{otherwise.} \end{cases}$$

Inverse witness for correctness. The EA also introduces an auxiliary polynomial $\mathsf{Inv}(X)$ and enforces, for all $X \in H$,

$$(\mathsf{Code}(X) - c) \cdot \mathsf{Inv}(X) - (1 - \mathsf{Eq}(X)) = 0,$$

together with the booleanity constraint

$$\mathsf{Eq}(X)\big(\mathsf{Eq}(X) - 1\big) = 0.$$

Intuition: if $\mathsf{Code}(X) \neq c$ then $\mathsf{Eq}(X) = 0$ and $\mathsf{Inv}(X) = (\mathsf{Code}(X) - c)^{-1}$ exists; if $\mathsf{Code}(X) = c$ then $\mathsf{Eq}(X) = 1$ and the equation forces $0 = 0$ without requiring an inverse.

Non-membership on ballot b. Finally, the EA enforces that no equality occurs inside the ballot block:

$$\sum_{i=0}^{n-1} \mathsf{Sel}_b(\omega^i) \cdot \mathsf{Eq}(\omega^i) = 0.$$

(Equivalently, define $\mathsf{Hit}(X) = \mathsf{Sel}_b(X) \cdot \mathsf{Eq}(X)$ and prove its sum is 0.)

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